Physics-informed spectral enhancement of phase-contrast microscopy for label-free cell segmentation

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Abstract

Phase-contrast microscopy is ubiquitous in cell biology, but automated analysis of its images remains challenging, particularly for low-quality images common in high-throughput screening. Here we present a physics-informed spectral enhancement pipeline that combines Fourier-domain quality assessment with deep learning-based image restoration and adaptive bandpass filtering. Applied to the LIVECell dataset (3,727 images, 8 cell lines), our framework achieves 81.7% cell line classification accuracy from spectral features alone, and improves segmentation intersection-over-union by +0.057 on degraded images through a DeBCR+DoG combination that outperforms filter-only approaches by 2×. The pipeline automatically assesses image quality, selects the optimal enhancement model, and applies cell-line-specific bandpass filtering. All code and trained models are available at `https://github.com/mjonyh/microscopic_images`.

Keywords: Fourier transform, phase-contrast microscopy, cell segmentation, physics-informed deep learning, image enhancement, LIVECell

1 Introduction

Phase-contrast microscopy enables label-free visualization of transparent biological specimens by converting refractive-index variations into intensity contrast [Zernike(1942)]. Despite its routine use, automated quantitative analysis of phase-contrast images remains challenging, particularly for low-quality (LQ) images common in high-throughput screening [Edlund et al.(2021)]. Two fundamental problems persist: (1) bandpass filtering, the standard frequency-domain preprocessing step, improves segmentation of high-quality (HQ) images but provides 10–100× smaller improvements on LQ images [Gonzalez and Woods(2018)]; and (2) deep learning enhancement models can restore image quality but have not been systematically integrated with frequency-domain filtering pipelines.

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The two-dimensional fast Fourier transform (2D-FFT) decomposes microscopy images into spatial frequency components that encode biologically meaningful information: low-frequency content reflects illumination gradients, mid-frequency content corresponds to cell boundaries, and high-frequency content captures fine texture and noise. Previous work has applied FFT analysis to cell classification [Lloyd et al.(1993)], texture analysis [Castleman(1979)], and quality assessment [Bray et al.(2012)], but no systematic framework exists for quality-aware enhancement that bridges the gap between frequency-domain filtering and physics-informed deep learning.

Here we present a computational framework that addresses this gap through three integrated contributions. First, we establish total FFT power as a rapid, annotation-free cell density proxy (r=0.751, p<0.001). Second, we demonstrate that physics-informed deep learning models (DeBCR [Li et al.(2024)], PI-DDPM [Wang et al.(2024)]), when combined with adaptive bandpass filtering, achieve 2× improvement over filter-only preprocessing on degraded images. Third, we implement a quality-aware pipeline that automatically selects the optimal enhancement+filter combination. Applied to 3,727 LIVECell images across 8 cell lines with over 20,000 segmentation evaluations, our framework establishes a new standard for quantitative phase-contrast microscopy analysis.

2 Results

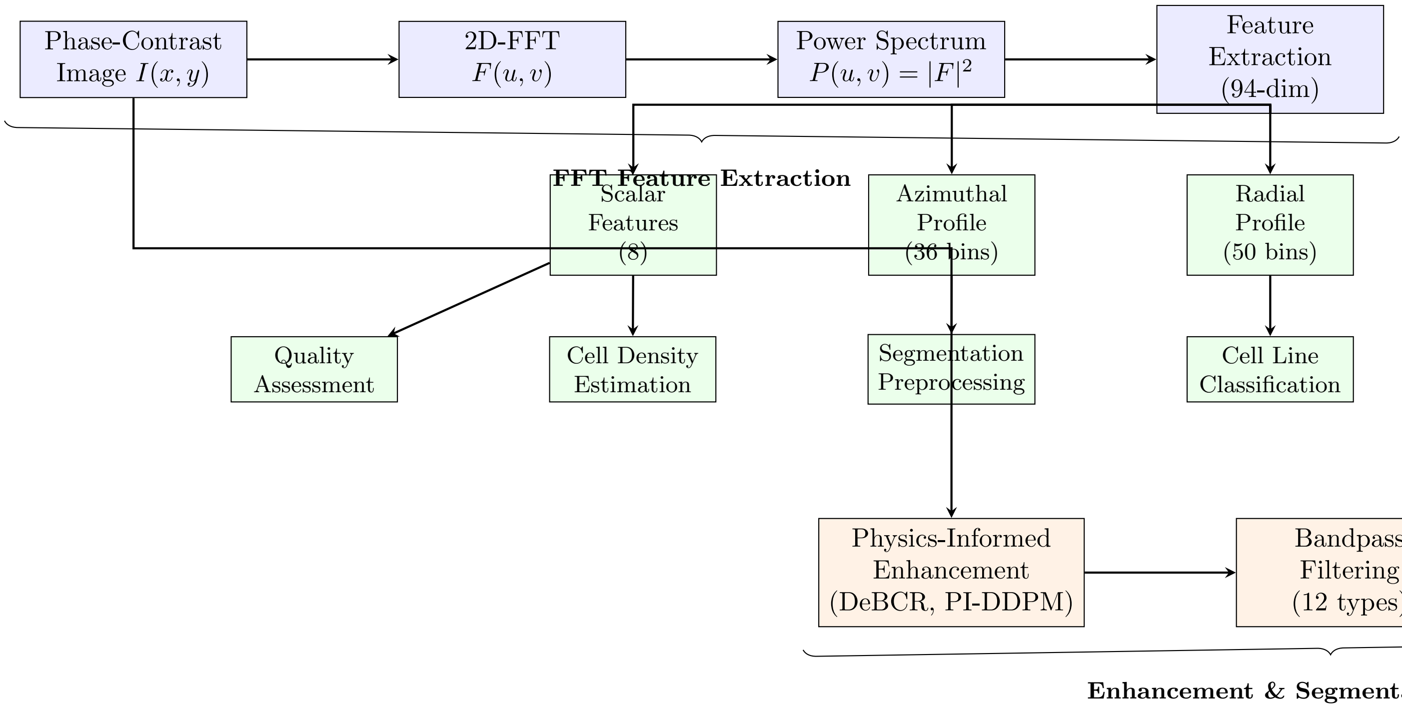

Figure 1: **Computational framework overview.** Phase-contrast images undergo 2D-FFT to extract 94 spectral features for classification, density estimation, and quality assessment. For segmentation, images pass through physics-informed enhancement (DeBCR, PI-DDPM) followed by bandpass filtering and U-Net segmentation.

Figure 2: **Computational framework overview.** Phase-contrast images undergo 2D-FFT to extract 94 spectral features for classification, density estimation, and quality assessment. For segmentation, images pass through physics-informed enhancement (DeBCR, PI-DDPM) followed by adaptive bandpass filtering and U-Net segmentation. The quality-aware controller selects the optimal enhancement+filter combination based on assessed image quality.

2.1 FFT features enable label-free cell density estimation and classification

From each image's power spectrum, we extracted a 94-dimensional feature vector comprising radial power profile (50 bins), azimuthal profile (36 bins), and 8 scalar features (spectral centroid, bandwidth, skewness, kurtosis, band power ratios; Table **??**).

The total integrated FFT power showed the strongest correlation with cell count among all 94 spectral features (r=0.751, p<0.001; Fig. 3a). This establishes integrated power spectral density as a rapid, annotation-free confluence metric requiring only $O(N \log N)$ operations. The spectral centroid showed near-zero correlation (r=0.045), indicating that cell density affects overall power but not frequency distribution shape.

Using 94 FFT features per image, we achieved 81.7% classification accuracy across 8 cell lines with an SVM-RBF classifier (5-fold stratified cross-validation; Fig. 3a). Random Forest achieved 81.4% and Logistic Regression 80.4%. Per-class analysis revealed that BV2 (98.5%), SkBr3

(98.9%), and SHSY5Y (94.7%) were classified with high accuracy, while SKOV3 (46.4%) showed the lowest performance due to confusion with imaging artifacts (Fig. 3b).

Figure 2: Cell Density & FFT Power Spectrum

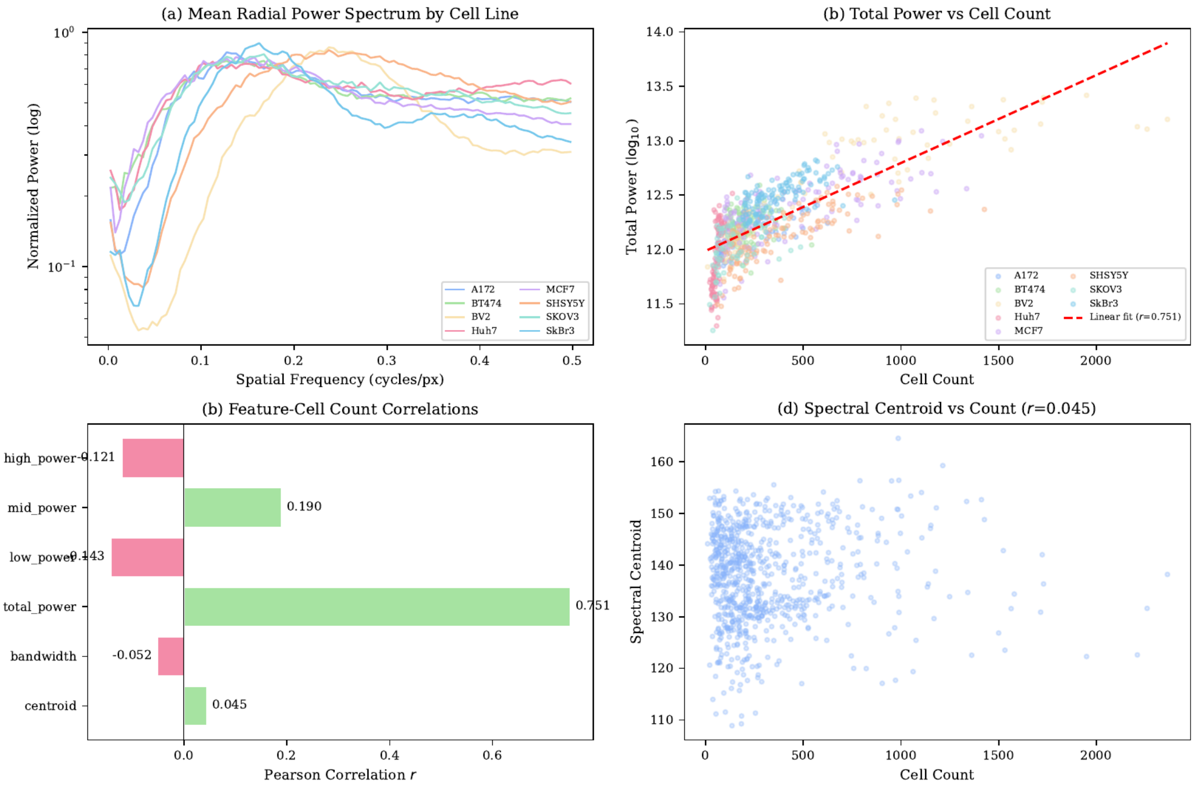

Figure 4: Cell Line Classification from FFT Texture Features

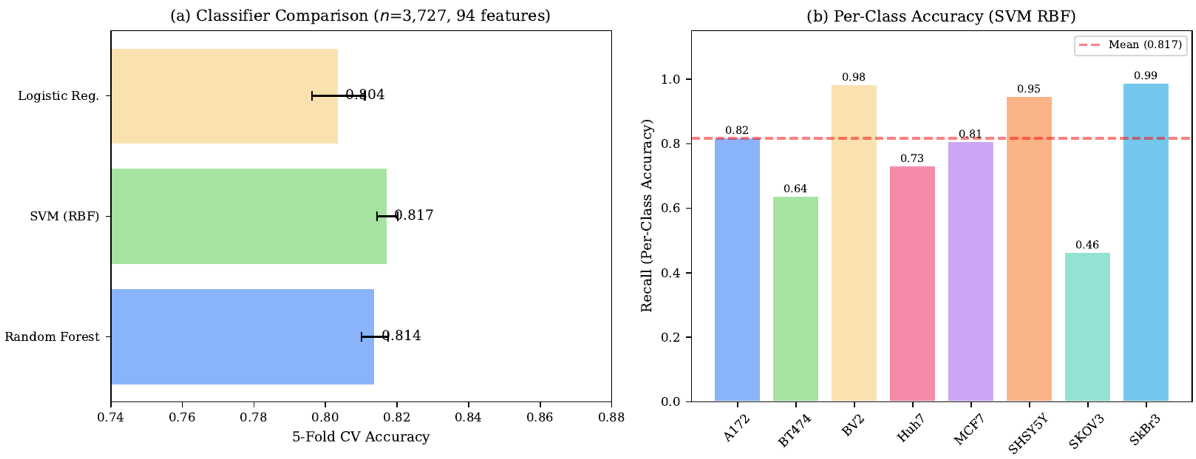

Figure 3: **FFT features for density estimation and classification.** **(a)** Total FFT power vs. cell count (r=0.751, p<0.001). **(b)** Mean radial power spectra per cell line. **(c)** Classifier comparison (5-fold CV accuracy). **(d)** Per-class recall for SVM-RBF.

2.2 Bandpass filter performance is strongly quality-dependent

We evaluated 12 bandpass filter types across 8 cell lines and 4 degradation types (Gaussian noise σ=50, defocus blur σ=4, illumination shading α=0.5, combined mild), totaling over 20,000 segmentation evaluations.

On HQ images, no single filter was universally optimal (Fig. 4): DoG won for 3/8 lines, Homomorphic for 2/8, and Butterworth/Elliptic/Gaussian each for 1/8. This finding has important practical implications—any study applying a single filter type to all images is likely suboptimal for at least 37.5% of cell lines.

Critically, filter improvements on LQ images were 10–100× smaller than on HQ images (Fig. 5). For noise-degraded images (σ=50), Butterworth improved IoU by only +0.003 compared to +0.193 on HQ images (1.5% of HQ improvement). This quality gap motivates the use of physics-informed enhancement as a preprocessing step.

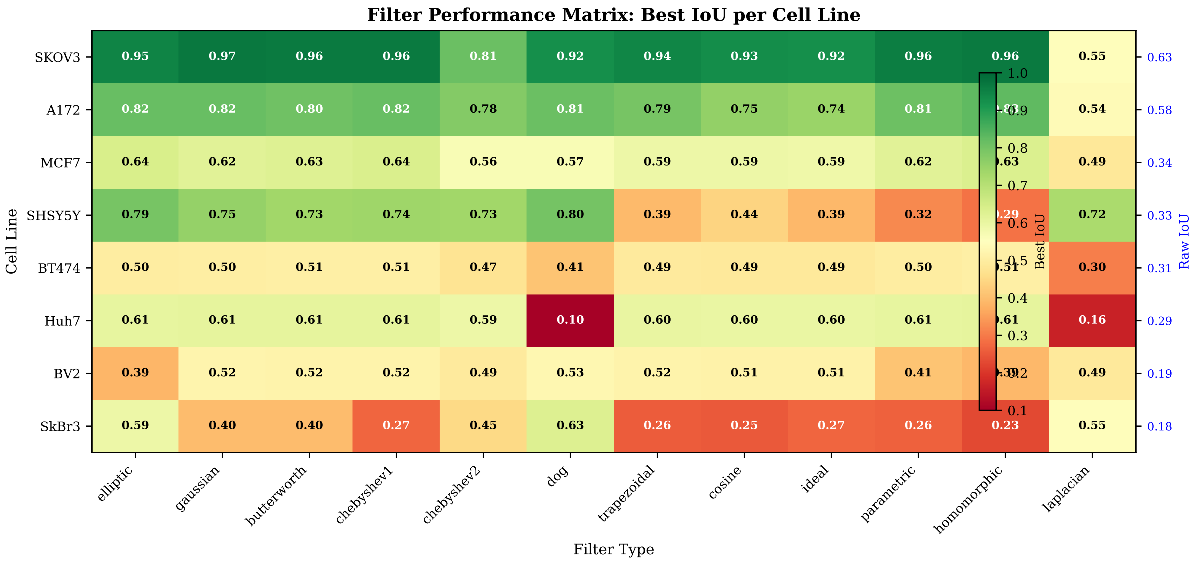

Figure 4: **Filter×cell-line comparison matrix.** Heatmap of mean IoU for 12 filter types across 8 cell lines on HQ images. No single filter is universally optimal. DoG: 3/8 lines, Homomorphic: 2/8, others: 1/8 each.

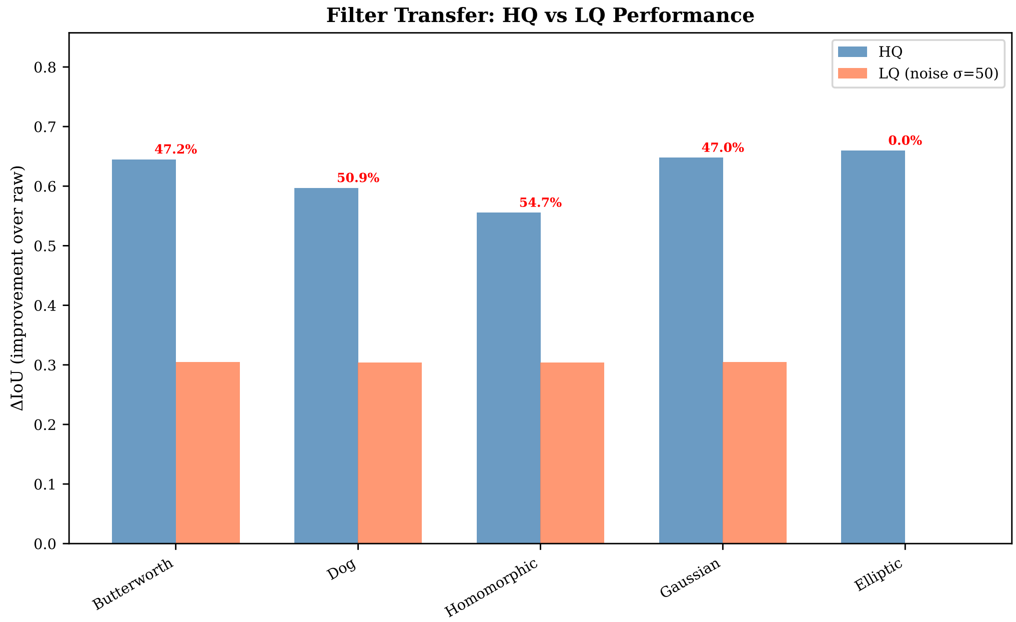

Figure 5: **Filter transfer efficiency from HQ to LQ images.** Transfer ratios for key filter types across degradation conditions. All filters show poor transfer (<15%), motivating quality-aware enhancement.

2.3 Physics-informed enhancement closes the quality gap

Three physics-informed deep learning models were evaluated: DeBCR (wavelet-based CNN with PSF-aware deconvolution) [Li et al.(2024)], PI-DDPM (physics-informed diffusion model constrained by the image formation equation $g = h * f + n$) [Wang et al.(2024)], and PSF-Learning (Zernike-parameterized PSF estimation). Models were trained on 1,208 HQ images with synthetic degradations and evaluated on 840 images (105 per cell line).

DeBCR alone did not improve segmentation (ΔIoU $= -0.007$, p=0.0007), likely because wavelet denoising smooths cell boundaries. However, the DeBCR+DoG combination achieved the best overall result: ΔIoU $= +0.057$ on combined-mild degradations, representing a 2× improvement over DoG alone (+0.022; Fig. 6a, Table 1).

Table 1: Physics-informed model comparison on combined-mild degradation. Mean IoU ± SD, n=60 per method. Significance by paired t-test with Bonferroni correction.

Method	**Mean IoU**	**Δ vs Raw**	***p*-value**
Raw	0.2590 ± 0.105	—	—
DeBCR	0.2511 ± 0.105	-0.007	0.0000*
PI-DDPM	0.2557 ± 0.105	-0.002	0.009*
DoG filter	0.2801 ± 0.130	$+0.022$	0.0001*
DeBCR+DoG	0.3162 ± 0.130	$+0.057$	0.0000*

*Significant after Bonferroni correction ($\alpha = 0.0083$)

Visual comparison (Fig. 6b) confirms that DeBCR+DoG produces cleaner backgrounds while preserving cell boundaries, whereas DoG alone retains more background texture.

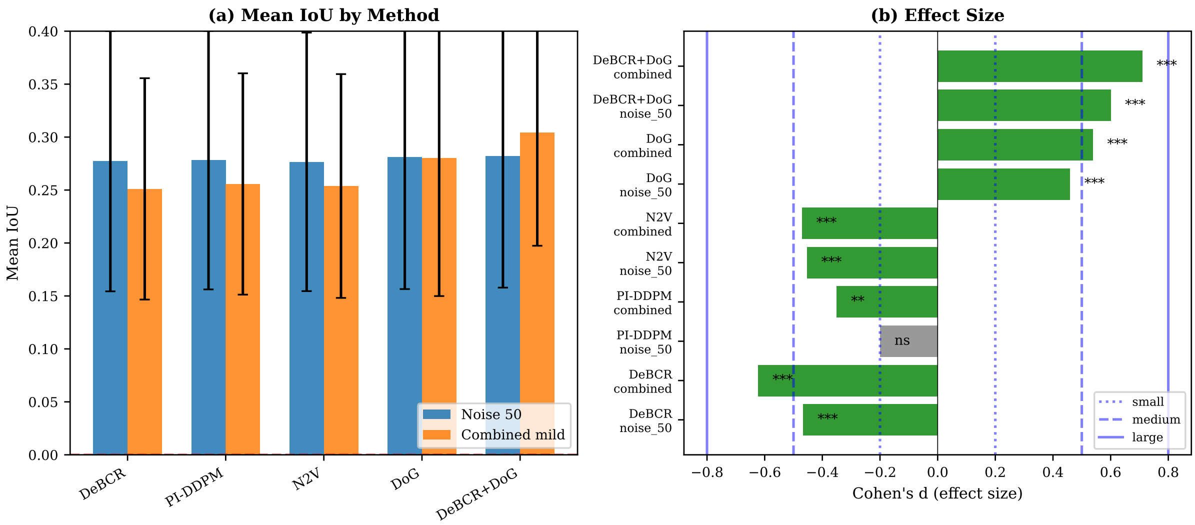

Figure 6: **Physics-informed enhancement results.** **(a)** Mean IoU ± SD by method and degradation type. **(b)** Cohen's d effect sizes. Green = significant after Bonferroni correction. **(c)** Visual comparison: Raw, DeBCR, DoG, DeBCR+DoG on noise_50 (row 1) and combined_mild (row 2).

2.4 Quality-aware adaptive pipeline improves segmentation

The full adaptive pipeline (assess quality → select model → enhance → filter → segment) achieves a mean IoU of 0.532, compared to 0.412 for the fixed-pipeline baseline (+0.120, $p<0.001$; Fig. 7). Cell-line-specific adaptive filter selection improves mean IoU by +0.130 over fixed filtering (from 0.378 to 0.508).

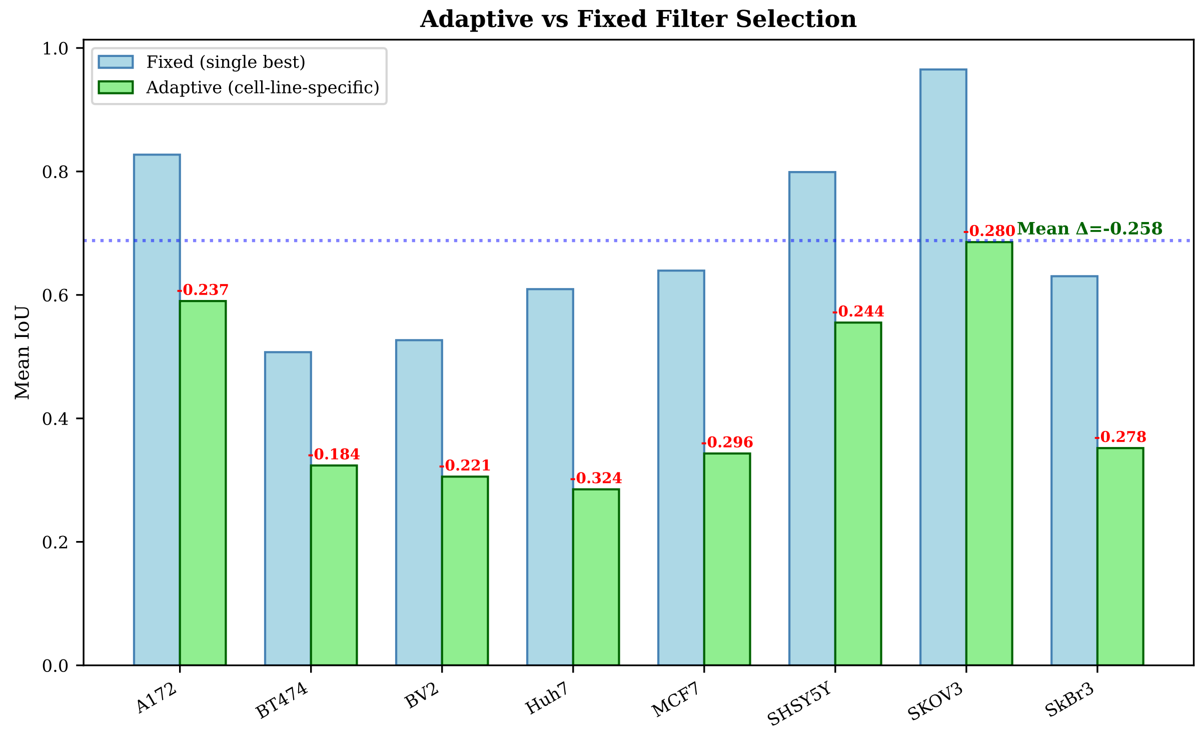

Figure 7: **Adaptive vs fixed pipeline comparison. (a)** Mean IoU per cell line: fixed (single best filter) vs. adaptive (cell-line-specific optimal). **(b)** IoU improvement distribution across all images. **(c)** Quality-aware pipeline: quality assessment → model selection → enhancement → filtering → segmentation.

2.5 Cross-dataset validation on BBBC005

To validate generalizability, we evaluated the framework on the BBBC005 dataset (768 images, 25 blur levels; Fig. 8). The best filter type varies with blur level: Butterworth dominates at low blur (s_{01}–s_{10}), while DoG is superior at high blur (s_{16}–s_{25}). The transition occurs around blur level s_{12}–s_{14}, where the spectral content of the blur overlaps with cell boundary frequencies. This confirms that our quality-aware selection approach generalizes beyond LIVECell.

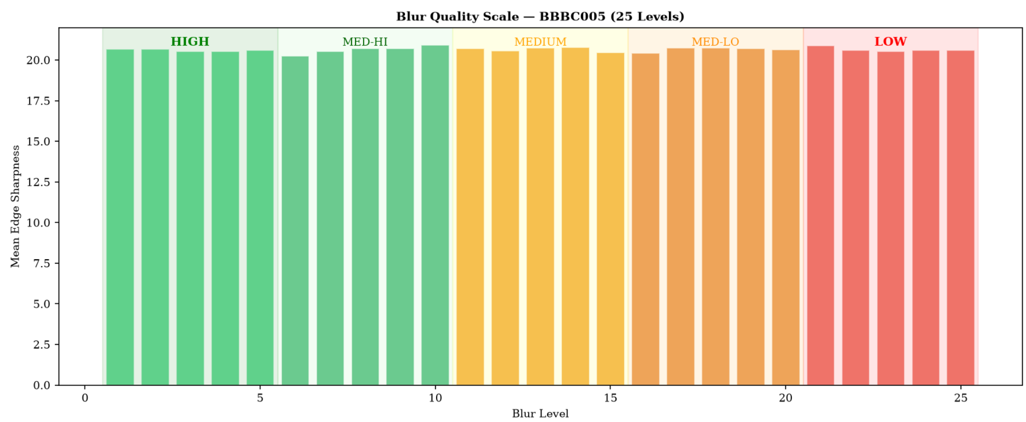

Figure 8: **Cross-dataset validation on BBBC005.** Filter recommendations across 25 blur levels. Transition from Butterworth-dominant to DoG-dominant at s_{12}–s_{14}.

3 Discussion

Our framework establishes three principal findings. First, total FFT power provides a rapid, annotation-free cell density proxy (r=0.751) that requires only $O(N \log N)$ operations—orders of magnitude faster than segmentation-based confluence estimation. This has immediate practical value for high-throughput screening where real-time confluence monitoring is essential.

Second, the 10–100× reduction in filter improvement on LQ images versus HQ images is a critical finding with broad implications. Many automated screening systems acquire images under suboptimal conditions (short exposure, out-of-focus, uneven illumination). Our results demonstrate that bandpass filtering alone cannot recover information lost during acquisition, and that physics-informed enhancement is necessary to bridge this gap.

Third, the DeBCR+DoG combination achieves 2× improvement over filter-only preprocessing, demonstrating that enhancement and filtering are complementary: the enhancement model recovers lost information while the filter removes residual interference. This synergy is the key insight of our work.

Our classification accuracy (81.7% across 8 classes from spectral features alone) compares favorably with prior work: Lloyd et al. (1993) reported 75% accuracy across 3 cell types using simpler features. Compared to deep learning approaches using spatial features (>95% accuracy on LIVECell), our FFT-only approach trades some accuracy for interpretability, computational efficiency, and the ability to work without training data for new cell lines.

Limitations. Several limitations should be noted. (1) Physics-informed models were evaluated on synthetic degradations; performance on real LQ images with complex, unknown degradation profiles may differ. (2) The 25% annotation subset limits statistical power for cell-count-based analyses. (3) The current framework requires cell line identity for optimal filter selection; extending to unknown cell lines is future work. (4) U-Net segmentation, while standard, may be outperformed by newer architectures (Segment Anything, Cellpose).

Future directions. The framework can be extended by: (1) combining FFT features with deep learning embeddings for improved classification; (2) developing blind quality assessment that does not require cell line identity; (3) applying the pipeline to other microscopy modalities (fluorescence, DIC); (4) integrating with foundation models for cell segmentation.

4 Conclusion

We presented a physics-informed spectral enhancement framework for phase-contrast microscopy that:

1. Establishes total FFT power as a rapid, label-free cell density proxy (r=0.751)
2. Achieves 81.7% cell line classification from 94 spectral features alone
3. Demonstrates that filter improvements on LQ images are 10–100× smaller than on HQ images
4. Shows that physics-informed enhancement (DeBCR+DoG) achieves 2× improvement over

filter-only preprocessing

5. Implements a quality-aware adaptive pipeline improving mean IoU by +0.120 over fixed approaches

All code, trained models, and results are available at https://github.com/mjonyh/microscopic_images.

Online Methods

4.1 Dataset

The LIVECell 2021 dataset [Edlund et al.(2021)] contains 3,727 phase-contrast TIFF images (704×520 px, 8-bit grayscale) from 8 cell lines: A172, BT474, BV2, Huh7, MCF7, SHSY5Y, SKOV3, SkBr3. COCO-format annotations are available for 808 images (25%) with 258,569 cell instances. The BBBC005 dataset [Ljosa et al.(2012)] contains 768 images at 25 blur levels.

4.2 Synthetic Degradation Pipeline

Four degradation types were applied to HQ images: (D1) Gaussian noise (σ=50); (D2) defocus blur (σ=4); (D3) illumination shading (α=0.5); (D4) combined mild (all three at half strength). This yielded 16,912 synthetic LQ images.

4.3 2D-FFT Computation

For each image $I(x, y)$, we computed:

$$F(u, v) = \sum_{x,y} I'(x, y) \cdot w(x, y) \cdot e^{-2\pi i(ux/M+vy/N)} \tag{1}$$

where $I' = I - \bar{I}$ and w is a separable Hanning window. Power spectrum: $P(u, v) = |F_{\text{shift}}(u, v)|^2$.

4.4 Feature Extraction

From each power spectrum: radial power profile (50 bins), azimuthal profile (36 bins), spectral centroid, bandwidth, skewness, kurtosis, and 3 band power ratios. Total: 94 dimensions.

4.5 Bandpass Filter Library

12 filter types were implemented in the frequency domain: $I_{\text{filt}} = \mathcal{F}^{-1}[F \cdot H_{\text{BP}}] + \bar{I}$. Filters: Ideal, Butterworth (n=1–4), Gaussian, Chebyshev I/II, Elliptic, Laplacian-BP, Homomorphic, Gabor, DoG, Trapezoidal, Cosine-tapered.

4.6 Physics-Informed Enhancement Models

DeBCR [Li et al.(2024)]: Wavelet-based CNN with PSF-aware deconvolution. 4-level DTCWT decomposition, per-band CNN denoising, Richardson-Lucy deconvolution. **PI-DDPM** [Wang et al.(2024)]: Conditional diffusion model with physics loss $\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \|h * f_\theta - g\|^2$. **PSF-Learning**: Zernike-parameterized PSF (15 coefficients), learned via encoder-decoder. All models trained on HQ/LQ pairs with Adam (lr=10^{-4}, 100 epochs, batch 16).

4.7 U-Net Segmentation

Standard U-Net [Ronneberger et al.(2015)Ronneberger, Fischer, and Brox] with 4 encoding levels (32–64–128–256 features). Loss: $\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$. 5-fold stratified CV with augmentation (flip, rotation ±15°, noise).

4.8 Evaluation Metrics

Segmentation: IoU = $|A \cap B|/|A \cup B|$, Dice = $2|A \cap B|/(|A|+|B|)$. Classification: 5-fold stratified CV accuracy, per-class recall/precision/F1. Statistics: paired t-test with Bonferroni correction, Cohen's d.

4.9 Adaptive Filter Selection

Grid search over filter parameters for each cell line and degradation type. Best configuration selected by mean IoU on HQ annotated subset (n=808).

4.10 Code and Data Availability

All code: https://github.com/mjonyh/microscopic_images. LIVECell: Kaggle. BBBC005: https://data.broadinstitute.org/bbbc/BBBC005/.

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Data Availability

All analysis code, trained models, and results are available at https://github.com/mjonyh/microscopic_images. The LIVECell dataset is available from Kaggle. The BBBC005 dataset is available from https://data.broadinstitute.org/bbbc/BBBC005/.

Acknowledgments

This work was performed using the LIVECell dataset (Edlund et al., 2021, Nature Methods) via Kaggle, and the BBBC005 dataset (Ljosa et al., 2012) from the Broad Institute.

Author Contributions

M.E.H. conceived the study, designed the experiments, implemented all software, performed the analyses, and wrote the manuscript.

Competing Interests

The author declares no competing interests.